



C23-AI/AIM/CAI/CCB/CCN/CIOT/CM-301

23102

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DCME-THIRD SEMESTER EXAMINATION

ENGINEERING MATHEMATICS—II

Time : 3 Hours ]

[ Total Marks : 80

**PART—A**

3×10=30

**Instructions :** (1) Answer **all** questions.

(2) Each question carries **three** marks.

1. Evaluate  $\int \left( 2\sec^2 x - e^x - \frac{1}{\sqrt{1-x^2}} \right) dx$

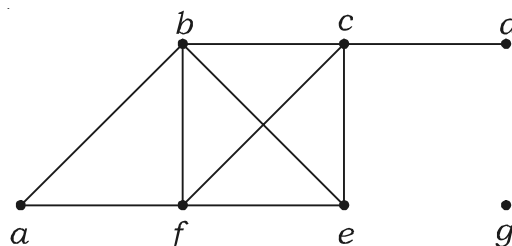
2. Evaluate  $\int x \sin(x^2 + 1) dx$

3. Evaluate  $\int_1^{\sqrt{3}} \frac{1}{1+x^2} dx$

4. Evaluate  $\int_0^{\frac{\pi}{2}} \sin x dx$

5. Find the differential equation of family of curves  $y = a \sin 3x$  where  $a$  is an arbitrary constant.

6. Find the degrees of vertices in the following graph :



7. Draw the graph with three vertices whose adjacency matrix is  $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ .
8. A bag containing 4 red balls, 6 black balls and 7 white balls. A ball is drawn at random. Find the probability that it is red ball or a black ball.
9. Find  $P(B / A)$ , if  $P(A \cap B) = \frac{2}{5}$ ,  $P(A) = \frac{2}{3}$ .
10. Find the Quartile Deviation of data 11, 6, 9, 21, 19, 17, 15.

### PART—B

10×5=50

**Instructions :** (1) Answer *any five* questions.  
(2) Each question carries **ten** marks.

11. (a) Evaluate  $\int \sin 4x \cos 3x \, dx$

(b) Evaluate  $\int \frac{3x+1}{(x-1)(x+3)} \, dx$

12. (a) Evaluate  $\int \tan^{-1} \left( \frac{2x}{1-x^2} \right) \, dx$

(b) Evaluate  $\int x^3 \cos x \, dx$

13. (a) Evaluate  $\int_0^1 \frac{\cos(\tan^{-1} x)}{1+x^3} \, dx$

(b) Show that  $\int_0^{\frac{\pi}{2}} \log \tan x \, dx = 0$

- 14.** (a) Solve the differential equation  $(1 + y^2)dx + (1 + x^2)dy = 0$ .
- (b) Solve the differential equation  $\frac{dy}{dx} + \frac{2y}{x} = \frac{1}{x^2}$ .
- 15.** (a) Two dies are thrown simultaneously. Find the probability that sum of on them is atmost 10.
- (b) Let  $A$  and  $B$  are two events with  $P(A) = \frac{1}{2}$ ,  $P(\bar{B}) = \frac{2}{3}$  and  $P(A \cap B) = \frac{1}{4}$ , find  $P(A / B)$  and  $P(B / A)$ .
- 16.** The chance that doctor  $A$  will diagnose a disease  $x$  correctly is 60%. The chance that a patient will die by his treatment after correct diagnosis is 40% and the chance of death of wrong diagnosis is 70%. A patient of doctor  $A$  who had disease  $x$  died. What is the chance that his disease was diagnosed correctly?
- 17.** Calculate the Spearman's coefficient of correlation and interpret the result between the marks in Statistics and Mathematics subjects for the following data :

|             |    |    |    |    |    |    |    |    |
|-------------|----|----|----|----|----|----|----|----|
| Statistics  | 65 | 66 | 67 | 67 | 68 | 69 | 70 | 72 |
| Mathematics | 67 | 68 | 65 | 68 | 72 | 72 | 69 | 71 |

**18.** Calculate the regression coefficients of  $Y$  on  $X$  and obtain the regression equation for the following data :

|     |   |   |   |   |   |   |    |    |
|-----|---|---|---|---|---|---|----|----|
| $X$ | 1 | 3 | 4 | 6 | 8 | 9 | 11 | 14 |
| $Y$ | 1 | 2 | 2 | 4 | 5 | 7 | 8  | 10 |

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